

Does Skill Abundance Still Matter?

The Evolution of Comparative Advantage in the 21st Century

Shin Kikuchi, UCSD

09/29/2026, Chicago Fed

Skill Abundance and Comparative Advantage

- Skill Abundance: Central to comparative advantage (**Heckscher-Ohlin**)
 - Skill-abundant countries specialize in skill-intensive sectors

“Illustrating” the Idea of Heckscher–Ohlin

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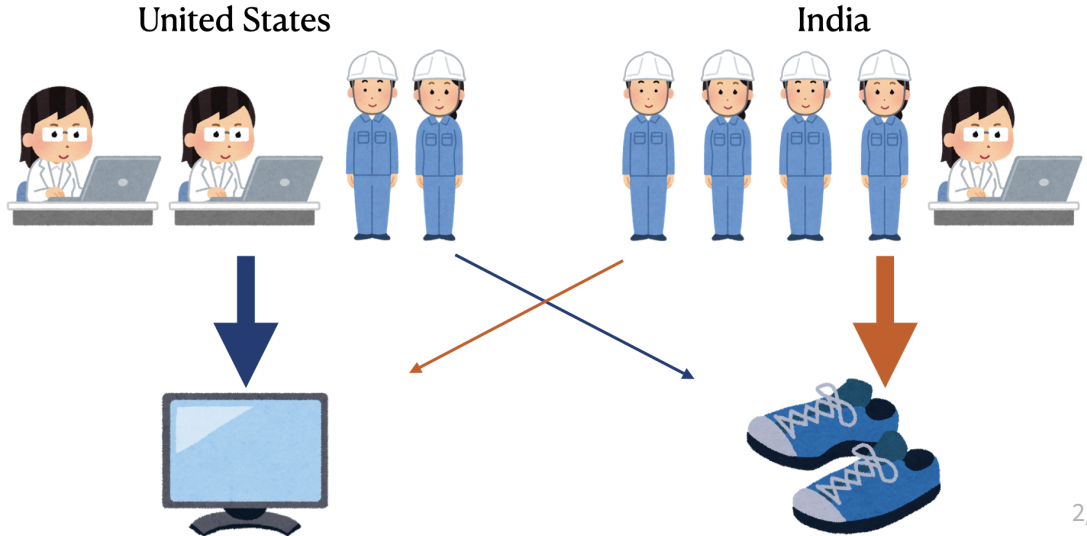
United States



India



“Illustrating” the Idea of Heckscher–Ohlin



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 - Implications for globalization, technology, and inequality

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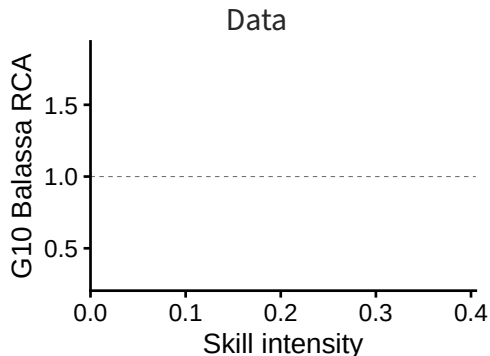
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- **This paper: What about the 21st century?**

At a Glance: Skill Abundance and Comparative Advantage

G10's Revealed Comparative Advantage (RCA) in skill-intensive sectors:

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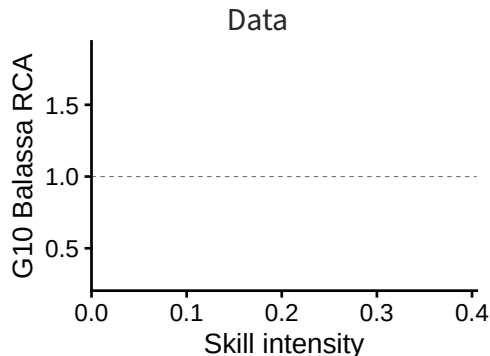
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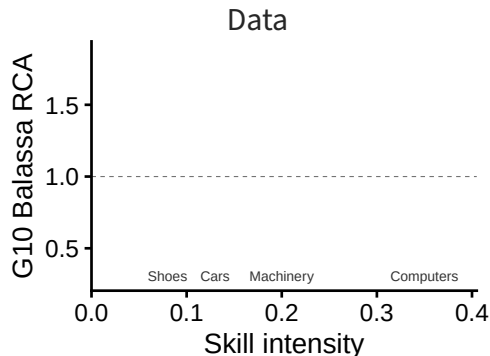


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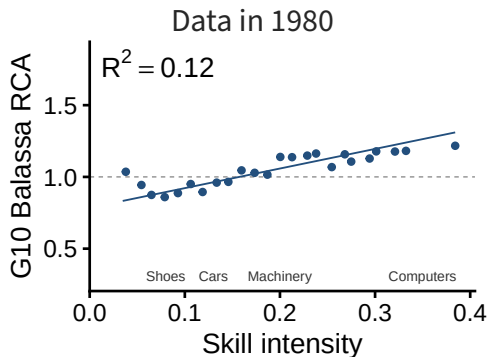
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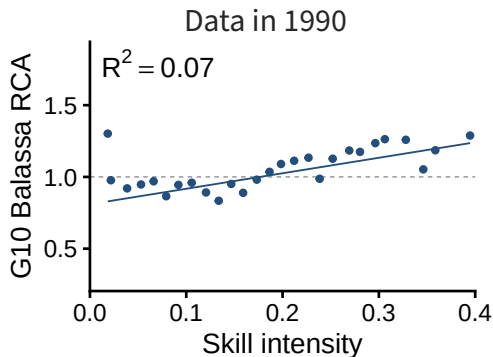
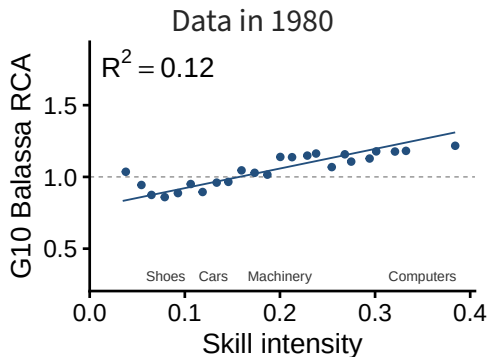
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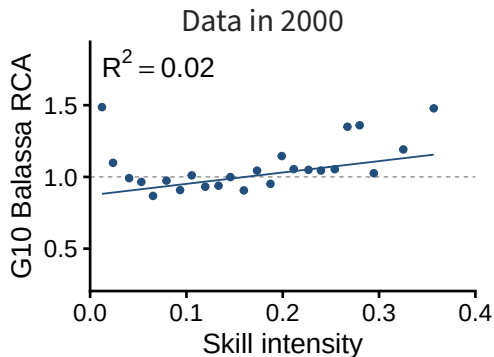
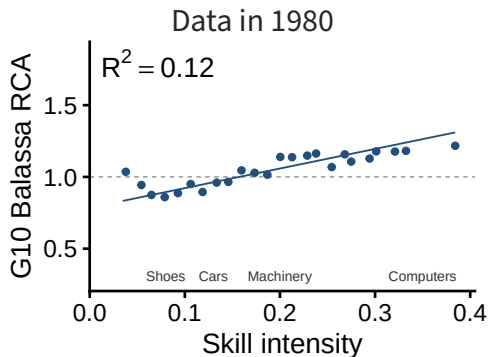
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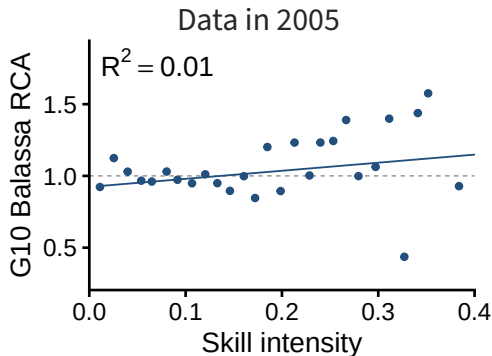
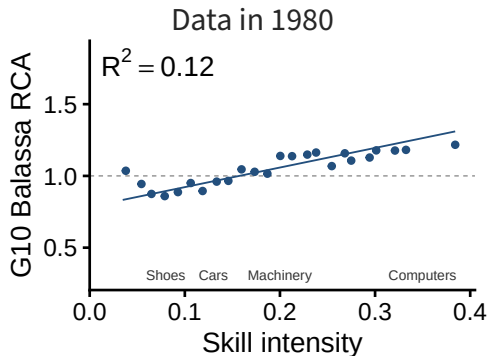
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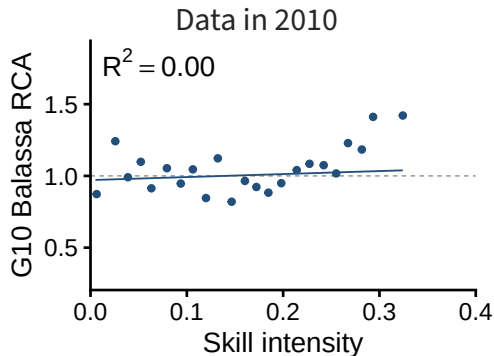
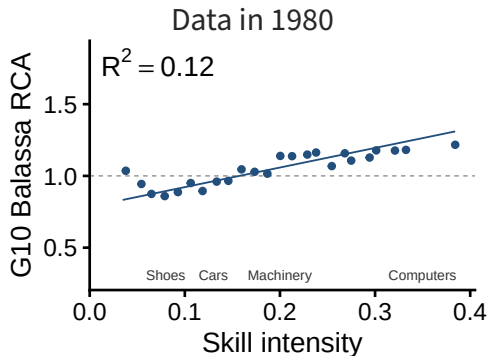
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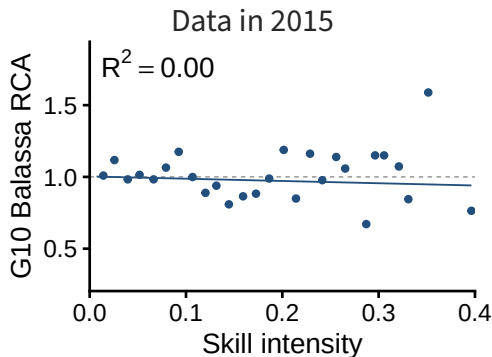
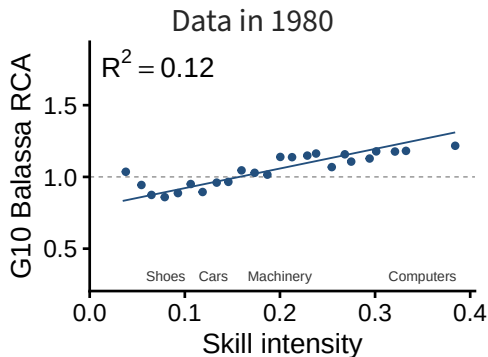
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 - **Lower trade costs amplify automation's effects**

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- **Ricardian:** MacDougall (1951), Stern (1962), Balassa (1963), Golub & Hsieh (2000), Nunn (2007), Levchenko (2007), Manova (2008), Costinot (2009), **Costinot et al. (2012)**
- **HO:** Leamer (1980, 1984), Bowen et al. (1987), Trefler (1993, 1995), Harrigan (1997), Davis & Weinstein (2001), Schott (2001), **Romalis (2004)**, Morrow (2010), **Chor (2010)**
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→ **Skill abundance mattered in the 1980s; not anymore after 2000.**

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 - **Interaction of technology and trade:** Xu (2001), Acemoglu (2003), Thoenig & Verdier (2003), Burstein et al. (2013), Parro (2013), Burstein & Vogel (2017), Morrow & Trefler (2022)
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→ **Automation ⇒ Comparative advantage and inequality**

Today's Plan

SKILL ABUNDANCE AS A SOURCE OF COMPARATIVE ADVANTAGE

Identifying Comparative Advantage: Refresher ▸ Relative Wage

- Multi-sector Eaton–Kortum model (Chor (2010), Costinot et al. (2012))

Exporter i , importer j , sector s : $\ln \text{Exports}_{i,j,s} = - \underbrace{\theta}_{\text{Trade Elas.}} \underbrace{\ln c_{i,s}}_{\text{Unit Cost}} + \underbrace{\tilde{\eta}_{i,j} + \tilde{\eta}_{j,s}}_{\text{FEs}}$

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Skill Abundance as a Source of Comparative Advantage

Do skill-abundant countries export more skill-intensive goods?

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- $\eta_{j,s}$: Importer-Sector FEs: control for tariffs, expenditure shares, ...

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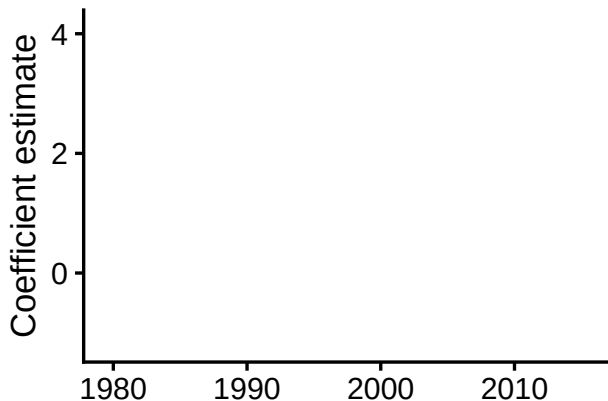
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- Poisson pseudo-maximum likelihood (PPML), separately by year
 - Includes zero trade flows
- Weights: exporter-sector total exports in 2015
 - Held fixed across destinations and years
- Standard errors clustered by exporter-sector

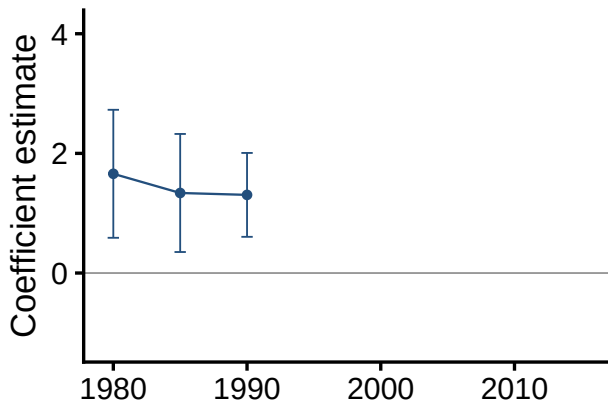
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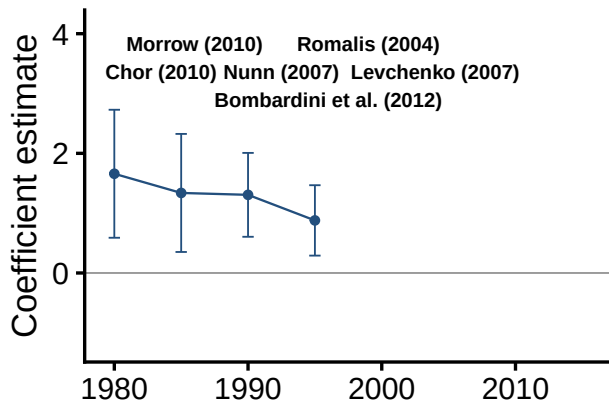
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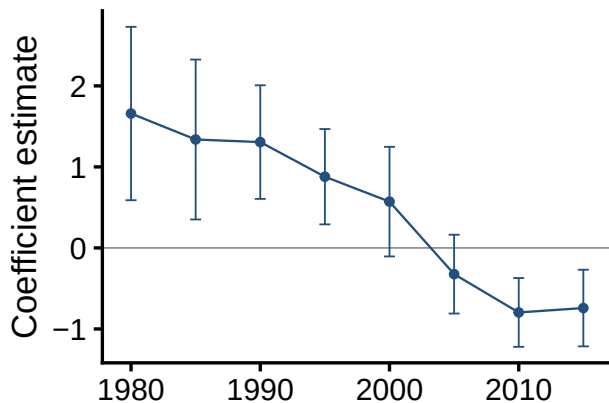
Early Results and Previous Research

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Declining Importance of Skill Abundance

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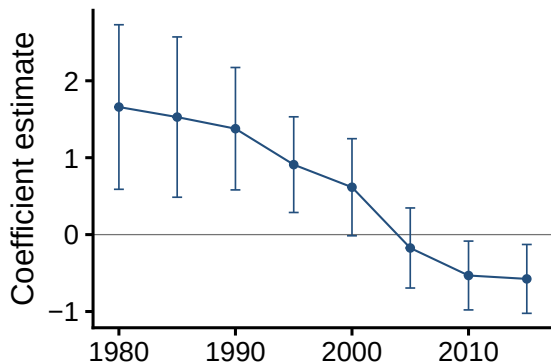


Fixing Skill Abundance and Intensity at 1980 Values

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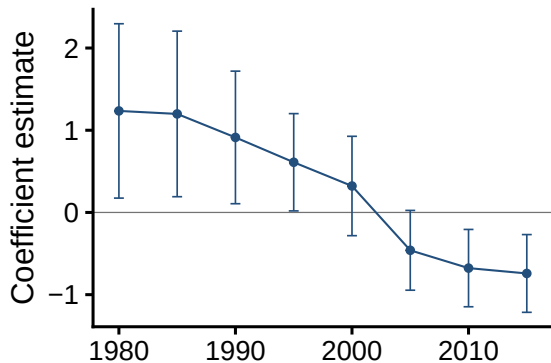


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More Robustness Checks

- Other sources of comparative advantage?
 - Capital [▸ go](#) Institutions [▸ go](#)
- Different specifications?
 - Log OLS [▸ go](#) Exporter-sector fixed effects [▸ go](#)
 - Total exports [▸ go](#)
- Different measures of skill abundance and intensity?
 - Chor (2010) [▸ go](#) Fixed inputs: 1980 [▸ go](#) 2015 [▸ go](#)

Empirics: Taking Stocks

- Skill abundance is used to be a source of comparative advantage
 - Interaction of skill abundance and intensity do not predict exports
- Not anymore after 2000

TRADE MODEL WITH AUTOMATION

Model: Overview

- Multi-sector Eaton–Kortum model with input-output linkages

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- Demand: Cobb–Douglas across sectors, CES (EK) within sectors, gravity
- Supply: Multiple production factors
 - Primary: labor H_{is} (high-skilled), L_{is} (low-skilled)
 - Additional inputs produced using sectoral outputs (roundabout)
 - ★ Automation capital: K_{is}
 - ★ Intermediates: $M_{i,i,r,s}$ (domestic), $M_{-i,i,r,s}$ (foreign, offshored)

Demand: Standard Multi-Sector Eaton–Kortum Model

- Country i, j , sector s
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- Gravity shares: G general use; M intermediate use

$$\pi_{ijs}^u = \frac{\lambda_{is}(c_{is}\tau_{ijs}^u)^{-\theta}}{\sum_{\ell \in \mathcal{I}} \lambda_{\ell s}(c_{\ell s}\tau_{\ell js}^u)^{-\theta}}$$

- Productivity scale λ_{is} : exogenous shifter
- Unit cost c_{is} : endogenously determined from production
- Iceberg trade costs: $\tau_{ijs}^G = \tau_{ijs}$, $\tau_{ijs}^M = \tau_{ijs}\chi_{ijs}$
- Trade elasticity $\theta > 0$

Supply 1/2: Task Framework in Production

- Eaton–Kortum with a task framework

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- Gross Output ($Z_{i,S}(\omega)$: Fréchet) for variety ω

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$$q_{is} = Z_{is} \cdot (H_{is})^{\alpha_{is}^H} \cdot (T_{is})^{\alpha_{is}^T} \cdot (M_{is})^{\alpha_{is}^M}.$$

- Cost shares: α_{is}^H high-skilled labor; α_{is}^T tasks
- α_{is}^M : intermediates; $\alpha_{is}^H + \alpha_{is}^T + \alpha_{is}^M = 1$

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- Task-specific productivities: $\psi_{is}^L(x)$ and $\psi_{is}^K(x)$

$$T_{is}(x) = \underbrace{\psi_{is}^L(x) \cdot L_{is}(x)}_{\text{Low-skilled labor}} + \underbrace{\psi_{is}^K(x) \cdot K_{is}(x)}_{\text{Capital}}$$

Supply 2/2: Task Allocation $\mathcal{T}_{is}^f$, Task Share Γ_{is}^f

- w_i^L : low-skilled wage; w_{is}^K : capital-service price
- Task-specific costs for $f \in \{L, K\}$:

$$\tilde{w}_{is}^L(x) = \frac{w_i^L}{\psi_{is}^L(x)}, \quad \tilde{w}_{is}^K(x) = \frac{w_{is}^K}{\psi_{is}^K(x)}$$

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- Cost minimization determines task sets and their shares:

$$\mathcal{T}_{is}^f = \{x : f = \operatorname{argmin}_{g \in \{L, K\}} \tilde{w}_{is}^g(x)\}, \quad \Gamma_{is}^f = \int_{\mathcal{T}_{is}^f} dx$$

Supply 2/2: Task Allocation $\mathcal{T}_{is}^f$, Task Share Γ_{is}^f

- Task allocation determines the task-composite price:

$$w_{is}^T = \exp \left(\int_{\mathcal{T}_{is}^L} \ln \tilde{w}_{is}^L(x) dx + \int_{\mathcal{T}_{is}^K} \ln \tilde{w}_{is}^K(x) dx \right)$$

- Task shares Γ_{is}^f set the domains in w_{is}^T
- w_{is}^M : intermediate-bundle price; Λ_{is} : cost-share constant
- Sectoral unit cost:

$$c_{is} = \Lambda_{is} (w_i^H)^{\alpha_{is}^H} (w_{is}^T)^{\alpha_{is}^T} (w_{is}^M)^{\alpha_{is}^M}$$

Equilibrium Conditions

Given factor endowments $\{H_i, L_i\}$, an equilibrium is a set of wages $\{w_i^H, w_i^L\}$

- Household and firm sourcing yields π_{ijs}^U from unit costs $\{c_{\ell s}\}$
- Unit cost c_{is} depends on $\{w_i^H, w_i^L\}$
 - $\{w_{is}^K, w_{is}^M\}$ also depend on wages through input-output linkages
- Goods and labor markets clear

DECOMPOSING CHANGES IN COMPARATIVE ADVANTAGE

Quantitative Data and Calibration

- **1980 benchmark; shocks through 2014**
 - 25 economies: 24 named economies and the rest of the world
 - 401 sectors: 390 manufacturing and 11 nonmanufacturing sectors

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 - Capital-flow tables: Ding (2026)
 - Fixed production shares and input-output weights

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- Capital-flow tables: Ding (2026)
- Fixed production shares and input-output weights

- **Elasticities**

- Trade elasticity $\theta = 4$; task-assignment elasticity $\varepsilon = 1.4$
- ε : elasticity of $\Gamma_{is}^K / \Gamma_{is}^L$ to w_i^L / w_{is}^K

Automation Shock

- **Equipment productivity from relative prices**

- NBER-CES, 1980–2014; 84 equipment sectors
- A hat denotes an endpoint/base ratio: $\hat{z} = z'/z$
- $\hat{\rho}_s$: equipment-price change relative to manufacturing prices

$$\hat{\lambda}_s = \hat{\rho}_s^{-\theta} \quad \text{in equipment sectors}$$

- **A sectoral shock common across countries**

- Identical $\hat{\lambda}_s$ in all 25 economies
- No direct Ricardian productivity effect on $\hat{\beta}$

Trade-Cost and Offshoring Shocks

Head and Ries (2001)'s "residual approach"

- Assume symmetric trade costs (and zero cost for domestic flows)
- Estimate bilateral trade costs for general use (final goods and capital) as

$$\tau_{ijs}^G = \left(\frac{X_{iis}^G \cdot X_{jjs}^G}{X_{ijs}^G \cdot X_{jis}^G} \right), \quad \tau_{ijs}^M = \left(\frac{X_{iis}^M \cdot X_{jjs}^M}{X_{ijs}^M \cdot X_{jis}^M} \right)$$

- Baseline trade cost changes $\hat{\tau}_{ijs}^G$ and additional offshoring wedge

$$\hat{\chi}_{ijs} \equiv \frac{\hat{\tau}_{ijs}^M}{\hat{\tau}_{ijs}^G}$$

Summary Statistics for the Calibrated Shocks

Summary Statistics (percentiles) for Shocks

Shock		p_{10}	p_{50}	p_{90}
Automation	$\hat{\rho}_s$	0.40	0.98	1.28
Trade costs	$\hat{\tau}_{ijs}$	0.30	0.66	1.25
Offshoring	$\hat{\chi}_{ijs}$	0.56	1.03	2.09
Endowment	$\hat{H}_i/\hat{L}_i$	2.99	4.54	7.77

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- Automation: relative equipment-price changes in 84 sectors
- Trade-cost/offshoring percentiles exclude diagonal and unit cells

Empirics using the Model's Sample

Let's set an empirical target for this sample...

- **Model regression sample**

- 24 named economies; cross-border manufacturing trade
- Hong Kong combined with China; domestic flows and ROW excluded

Empirics using the Model's Sample

Let's set an empirical target for this sample...

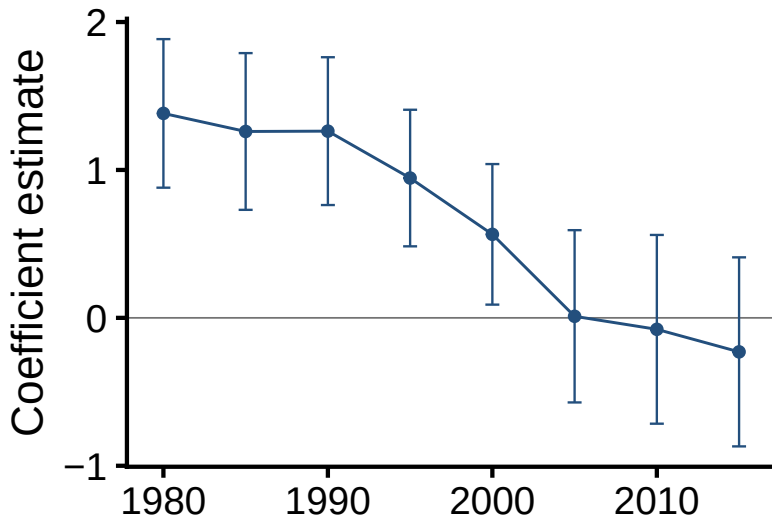
- **Model regression sample**

- 24 named economies; cross-border manufacturing trade
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- **Model regression specification**

- Unweighted PPML; skill interaction fixed at 1980
- Model shocks: 1980–2014
- Empirical endpoint: 2015 trade wave, averaging 2014–2016

Empirics using the Model's Sample



Decomposition via Counterfactual Combinations

- **Three shock blocks**

- Automation, trade costs, and offshoring
- Solve all 2^3 combinations from the 1980 equilibrium
 - ★ Shapley decomposition (invariant across decomp. orders)

Decomposition via Counterfactual Combinations

- **Three shock blocks**

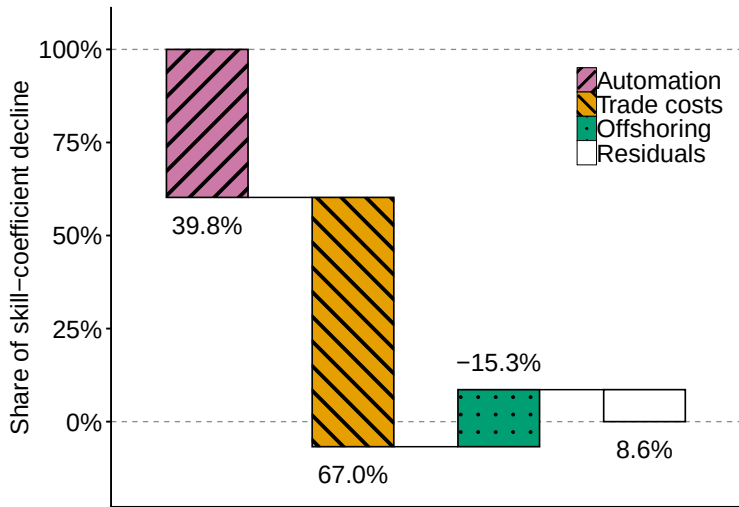
- Automation, trade costs, and offshoring
- Solve all 2^3 combinations from the 1980 equilibrium
 - ★ Shapley decomposition (invariant across decomp. orders)

- **Counterfactual regressions**

- Endowments and deficit shares remain fixed
 - ★ RHS of the regression is fixed
- Re-estimate PPML on each counterfactual trade matrix

$$\mathbb{E}[X_{i,j,s,t}^{Model}] = \exp\left(\beta_t \left[\alpha_{s,1980}^H \ln(H_{i,1980}/L_{i,1980})\right] + \eta_{i,j,t} + \eta_{j,s,t}\right)$$

Decomposing Changes in Comparative Advantage



Channel Accounting: From Shocks to the Coefficient

- E_{js}^u : use- u spending; $X_{ijs}^u = \pi_{ijs}^u E_{js}^u$; PPML uses $X_{ijs}^G + X_{ijs}^M$

Channel Accounting: From Shocks to the Coefficient

- E_{js}^u : use- u spending; $X_{ijs}^u = \pi_{ijs}^u E_{js}^u$; PPML uses $X_{ijs}^G + X_{ijs}^M$

$$d \ln X_{ijs}^u = \underbrace{D_{ijs}^u - \sum_{\ell} \pi_{\ell js}^u D_{\ell js}^u}_{\text{Direct}} - \underbrace{\theta \left(d \ln c_{is} - \sum_{\ell} \pi_{\ell js}^u d \ln c_{\ell s} \right)}_{\text{Unit cost}} + \underbrace{d \ln E_{js}^u}_{\text{Expenditure}},$$

where

- Direct: Efficiency term

$$D_{ijs}^u = d \ln \hat{\lambda}_s - \theta d \ln \tau_{ijs} - \theta \mathbf{1}\{u = M\} d \ln \chi_{ijs}$$

- Unit cost: Productivity, *net* of λ_{is} shifter.
- Expenditure: Expenditure patterns of destinations

Channel Accounting

Channel Accounting by Shocks: Each column reports one shock alone, 1980–2014.

	Trade costs	Automation	Offshoring
Direct	−0.98	0	−0.005
Unit cost	−0.19	−0.60	+0.20
Expenditure	+0.04	−0.19	+0.10
Total (one shock alone)	−1.14	−0.79	+0.29

- Trade costs: Direct (market access) accounts for 86.6%
- Automation: unit costs = task reassignment from low-skilled labor to capital

EFFECTS ON SKILL PREMIUM AND WELFARE

Macroeconomic Counterfactuals

- **Counterfactual assumptions**

- Measured 1980–2014 skill-supply changes in every experiment
- Intermediate-use wedges and deficit shares fixed at 1980

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- $(0, 0)$: endowments alone; SP: skill premium; W : welfare

Macroeconomic Counterfactuals

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- Measured 1980–2014 skill-supply changes in every experiment
- Intermediate-use wedges and deficit shares fixed at 1980

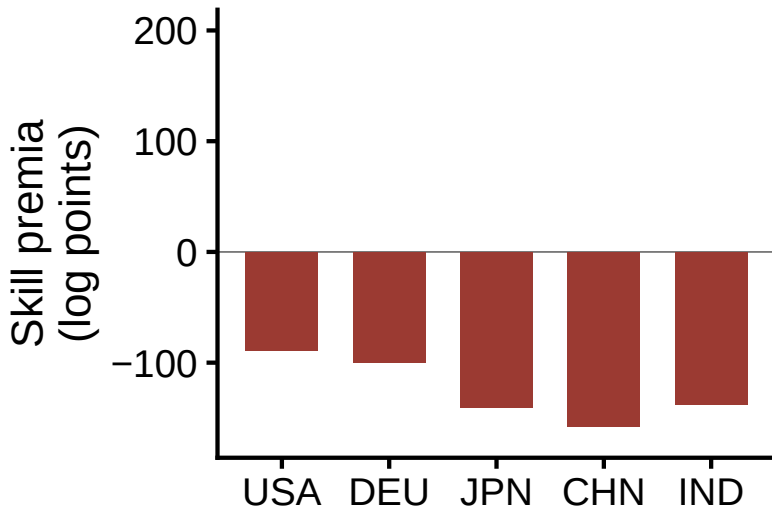
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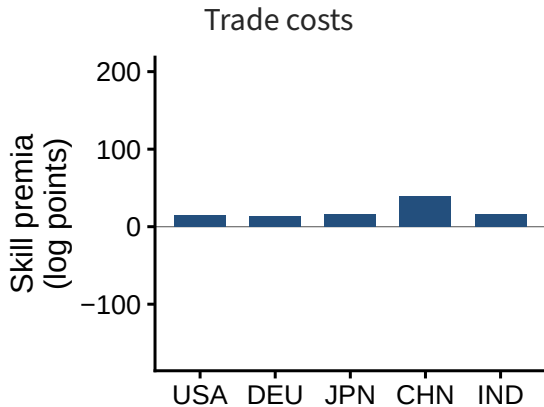
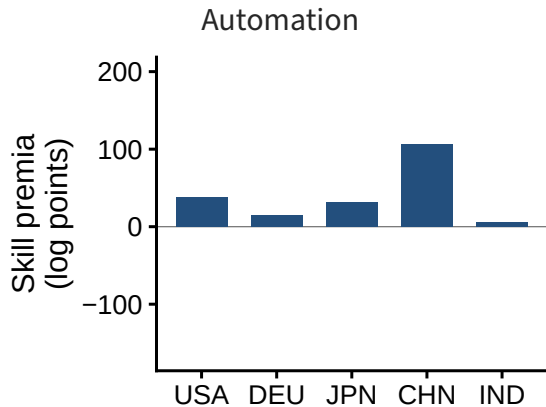
- **Shock effects** (y_i^Y : log-point outcome relative to 1980)

$$y_i^Y(a, t) - y_i^Y(0, 0), \quad Y \in \{\text{SP}, W\}$$

Endowment Changes Alone: Relative to 1980



The Race for Skills: Automation and Trade Costs



Both shocks raise the skill premium in every economy.

Task Reassignment and Sectoral Reallocation

- **Factor-market clearing**

- b_{is}^H, b_{is}^L : wage-bill shares; R_{is} : sectoral revenue
- Γ_{is}^L : low-skilled task share

$$d \ln \left(\frac{w_i^H}{w_i^L} \right) = \underbrace{d \ln \left(\frac{L_i}{H_i} \right)}_{\text{Relative supply}} + \underbrace{\sum_s (b_{is}^H - b_{is}^L) d \ln R_{is}}_{\text{Sectoral reallocation}} - \underbrace{\sum_s b_{is}^L d \ln \Gamma_{is}^L}_{\text{Task reassignment}}$$

Skill-Premium Accounting

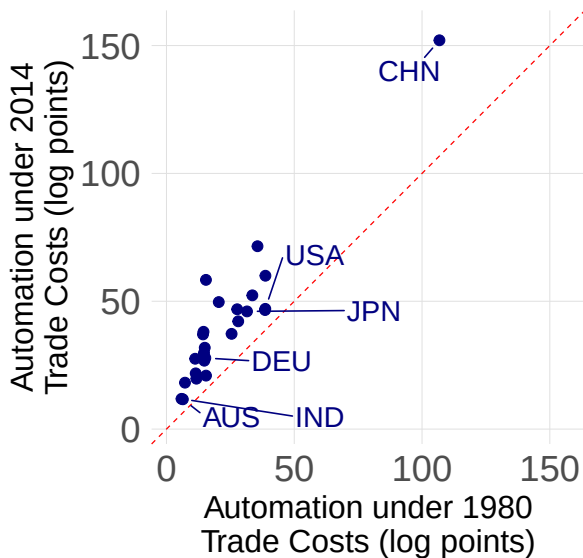
- **Mean effects across 25 economies; log points**

- First three: relative to endowments alone; last two: relative to 1980

Shocks	Supply	Sec. Reallocation	Task	Total
Automation	0.00	−2.49	26.43	23.94
Trade costs	0.00	−6.00	30.20	24.20
Both Shocks	0.00	−9.10	74.61	65.51
Endowment Only	−153.12	−1.34	15.81	−138.65
Both Shocks + Endowment	−153.12	−10.44	90.41	−73.15

- Task reassignment raises skill premia; sectoral reallocation partly offsets it

Lower Trade Costs Amplify Automation: Skill Premia



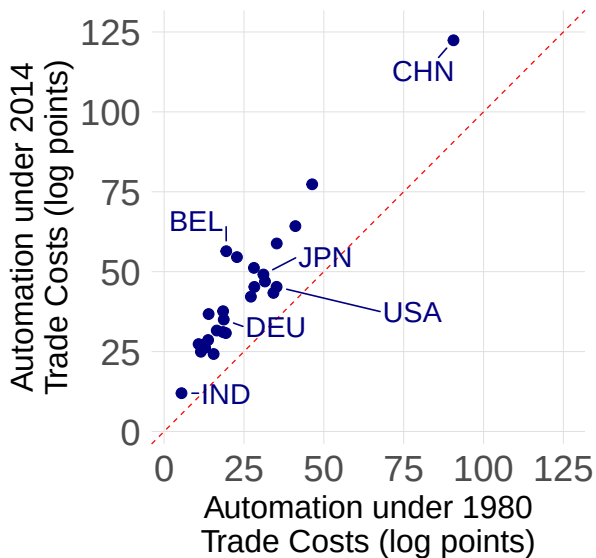
Welfare Effects

- **Across 25 economies.**

- First three: relative to endowments alone; last two: relative to 1980

Shocks	Min	Median	Max
Automation	5.4 (IND)	19.5	90.6 (CHN)
Trade costs	16.9 (AUS)	39.0	100.6 (BEL)
Both Shocks	29.3 (IND)	71.9	164.7 (CHN)
Endowment Only	7.1 (JPN)	39.7	126.4 (MEX)
Both Shocks + Endowment	80.1 (FIN)	132.4	242.0 (CHN)

Lower Trade Costs Amplify Automation: Welfare



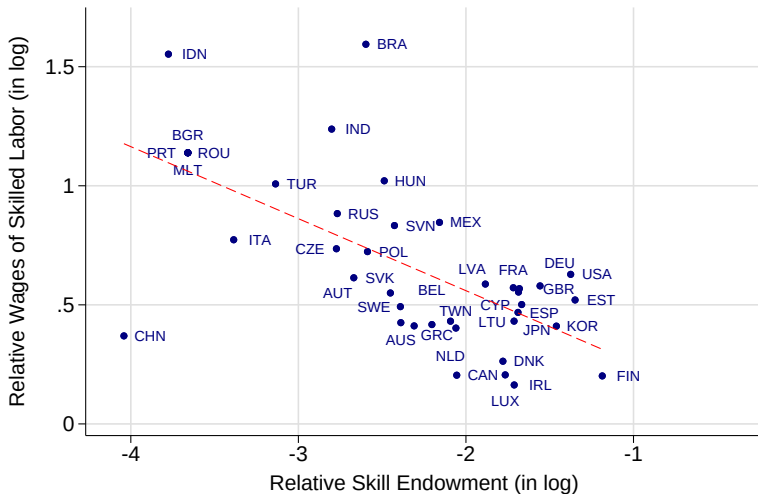
CONCLUSION

Conclusion

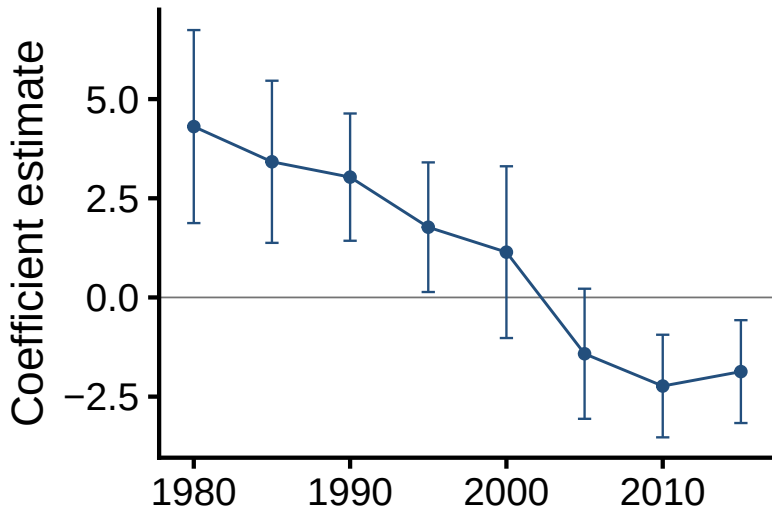
- **Skill abundance no longer predicts skill-intensive specialization**
 - Lower trade costs and automation explain most of the decline
 - Market access accounts for most of the trade-cost contribution
- **Lower trade costs amplify automation's effects**
 - Task reassignment raises skill premia
 - Welfare increases

APPENDIX

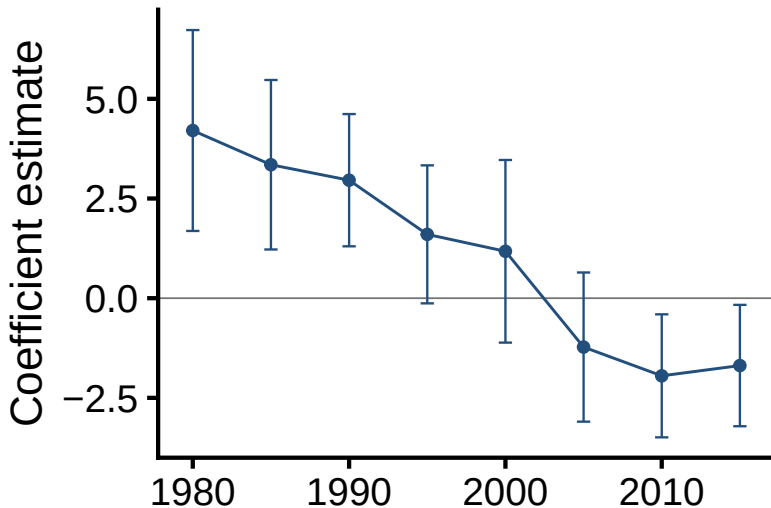
Relative Skilled Wages and Skill Endowment



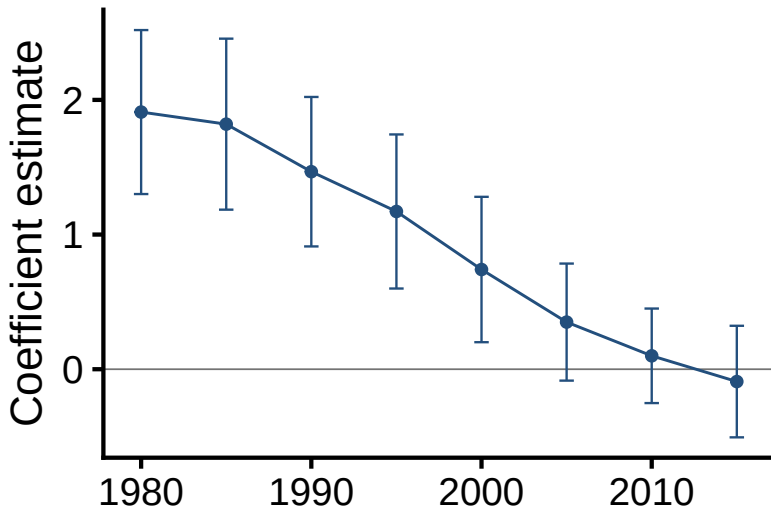
Controlling for Capital Intensity



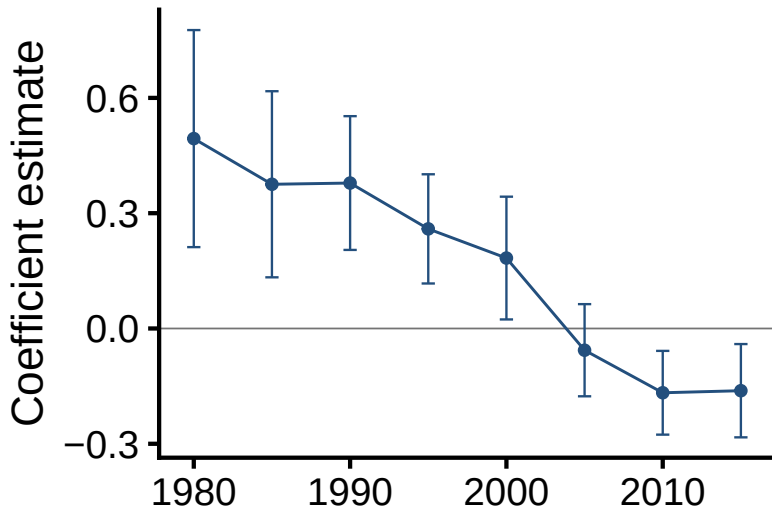
Controlling for Capital Intensity and Institutions



Log OLS Specification



Chor (2010): Log Non-Production / Production Employment



Model (1/7): Gross Production

- **Gross production**

- i : country; s : sector
- $H_{is}(\omega)$: high-skilled labor; $T_{is}(\omega)$: tasks; $M_{is}(\omega)$: intermediate inputs

$$q_{is}(\omega) = Z_{is}(\omega) \cdot [H_{is}(\omega)]^{\alpha_{is}^H} \cdot [T_{is}(\omega)]^{\alpha_{is}^T} \cdot [M_{is}(\omega)]^{\alpha_{is}^M}$$

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$$\alpha_{is}^H + \alpha_{is}^T + \alpha_{is}^M = 1$$

- **Fréchet productivity** (λ_{is} : scale; θ : trade elasticity)

$$\Pr[Z_{is}(\omega) \leq z] = \exp(-\lambda_{is}z^{-\theta})$$

Model (2/7): Tasks

- **Task composite**

- Tasks $x \in [0, 1]$ receive equal Cobb–Douglas weights

$$T_{is}(\omega) = \exp \left(\int_0^1 \ln T_{is}(\omega, x) dx \right)$$

Model (2/7): Tasks

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$$T_{is}(\omega) = \exp \left(\int_0^1 \ln T_{is}(\omega, x) dx \right)$$

- **Production of each task**

- $\psi_{is}^L(x)$ and $\psi_{is}^K(x)$: task-specific productivities

$$T_{is}(\omega, x) = \underbrace{\psi_{is}^L(x) \cdot L_{is}(\omega, x)}_{\text{Low-skilled labor}} + \underbrace{\psi_{is}^K(x) \cdot K_{is}(\omega, x)}_{\text{Capital}}$$

Model (3/7): Task Costs and Assignment

- **Task-specific unit costs**

- $w_i^{H,L}$: high-/low-skilled wages; w_{is}^K : capital-service price

$$\tilde{w}_{is}^L(x) = \frac{w_i^L}{\psi_{is}^L(x)} \quad \tilde{w}_{is}^K(x) = \frac{w_{is}^K}{\psi_{is}^K(x)}$$

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- **Automation**

- Task x goes to capital when

$$\frac{\psi_{is}^K(x)}{\psi_{is}^L(x)} > \frac{w_{is}^K}{w_i^L}$$

- Lower capital-service prices expand capital's task set

Model (4/7): Capital, Intermediate Inputs, and Unit Costs

- **Capital services/intermediate inputs**

- $P_{ir}^{G,M}$: general/intermediate prices; κ, γ : capital/input weights

$$w_{is}^K = \prod_r (P_{ir}^G)^{\kappa_{i,rs}}, \quad w_{is}^M = \prod_r (P_{ir}^M)^{\gamma_{i,rs}}$$

- **Task-composite price** ($a_{is}(x)$: minimum cost of task x)

$$w_{is}^T = \exp \left(\int_0^1 \ln a_{is}(x) dx \right)$$

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- **Sectoral unit cost** (Λ_{is} : production-share constant)

$$c_{is} = \Lambda_{is} (w_i^H)^{\alpha_{is}^H} (w_{is}^T)^{\alpha_{is}^T} (w_{is}^M)^{\alpha_{is}^M}$$

Model (5/7): General- and Intermediate-Use Sourcing

- **General use**

- π_{ijs}^G : exporter i 's share; τ_{ijs} : iceberg trade cost

$$\pi_{ijs}^G = \frac{\lambda_{is}(c_{is}\tau_{ijs})^{-\theta}}{\sum_{\ell} \lambda_{\ell s}(c_{\ell s}\tau_{\ell js})^{-\theta}}$$

Model (5/7): General- and Intermediate-Use Sourcing

- **General use**

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- **Intermediate use**

- χ_{ijs} : additional delivered-cost wedge

$$\pi_{ijs}^M = \frac{\lambda_{is}(c_{is}\tau_{ijs}\chi_{ijs})^{-\theta}}{\sum_{\ell} \lambda_{\ell s}(c_{\ell s}\tau_{\ell js}\chi_{\ell js})^{-\theta}}$$

Model (6/7): Trade Flows

- **Expenditure and trade**

- E_{js}^u : expenditure in country j , sector s , use u
- $u \in \{G, M\}$: general or intermediate use

$$\chi_{ijs}^u = \pi_{ijs}^u E_{js}^u$$

Model (6/7): Trade Flows

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- $u \in \{G, M\}$: general or intermediate use

$$X_{ijs}^u = \pi_{ijs}^u E_{js}^u$$

- **Total trade flows**

$$X_{ijs} = X_{ijs}^G + X_{ijs}^M$$

- General use: consumption and capital purchases
- Offshoring: foreign share of intermediate expenditure

Model (7/7): Household Income and Expenditure

- **Households**

- d_i : deficit share; V^W : world factor income
- Cobb–Douglas preferences; v_{is} : expenditure share

$$I_i = w_i^H H_i + w_i^L L_i + d_i V^W$$

Model (7/7): Household Income and Expenditure

- **Households**

- d_i : deficit share; V^W : world factor income
- Cobb–Douglas preferences; ν_{is} : expenditure share

$$I_i = w_i^H H_i + w_i^L L_i + d_i V^W$$

$$\underbrace{E_{is}^F}_{\text{Final}} = \nu_{is} I_i, \quad E_{is}^G = E_{is}^F + \underbrace{E_{is}^K}_{\text{Capital}}$$

Equilibrium

- **Equilibrium wages**

- Clear goods market

- ★ R_{is} : sales across destinations and uses

$$R_{is} = \sum_j \left(\pi_{ijs}^G E_{js}^G + \pi_{ijs}^M E_{js}^M \right)$$

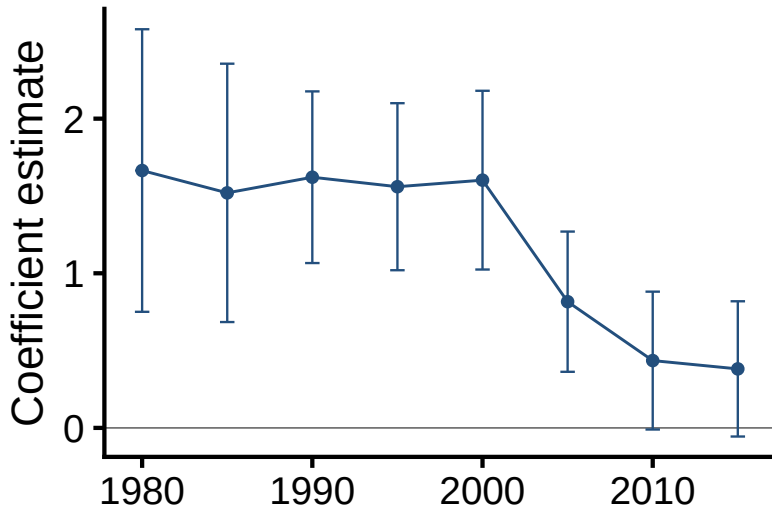
- Labor markets (Γ_{is}^L : low-skilled task share)

$$w_i^H H_i = \sum_s \alpha_{is}^H R_{is},$$

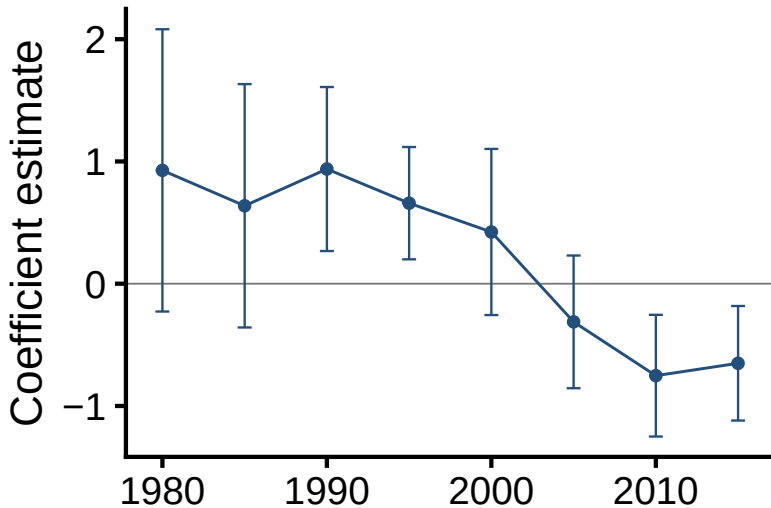
$$w_i^L L_i = \sum_s \alpha_{is}^T \Gamma_{is}^L R_{is}$$

- Determine prices, task shares, and trade flows

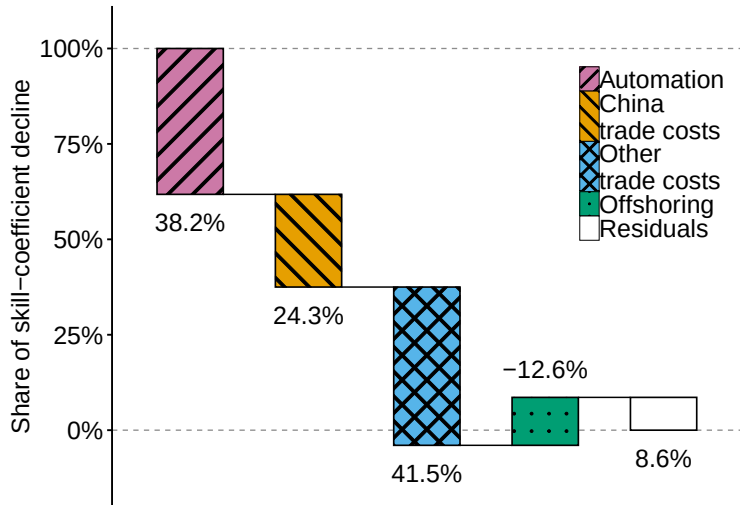
Exporter-Sector Fixed Effects



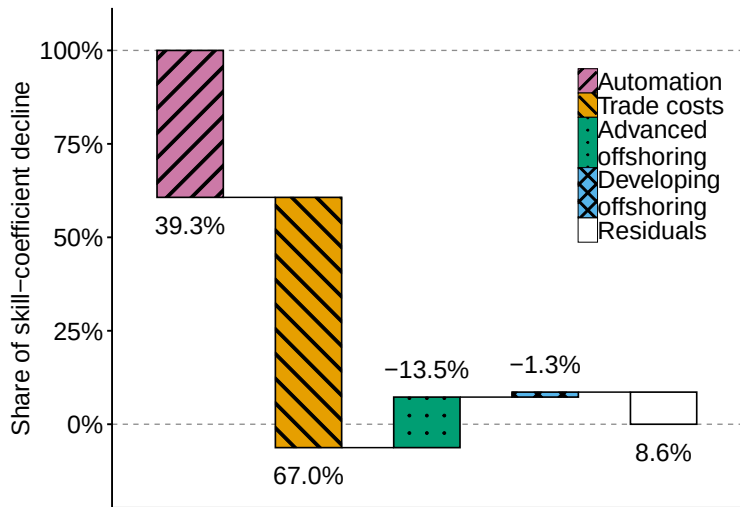
Romalis (2004): Total Exports



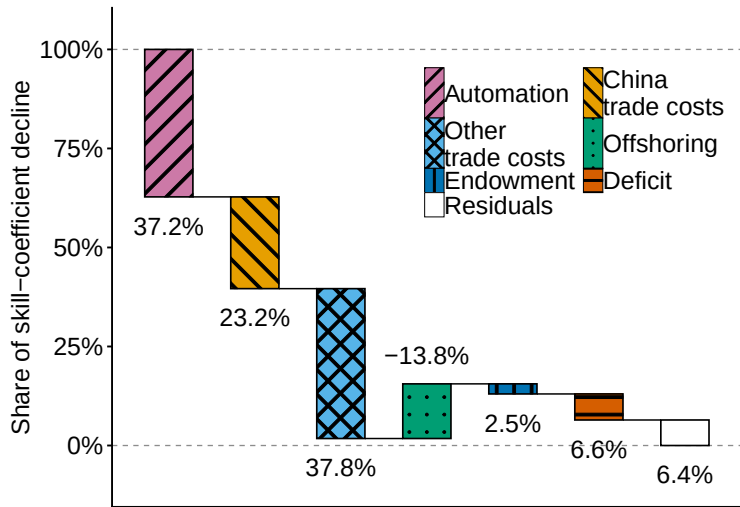
China and Other Trade Costs



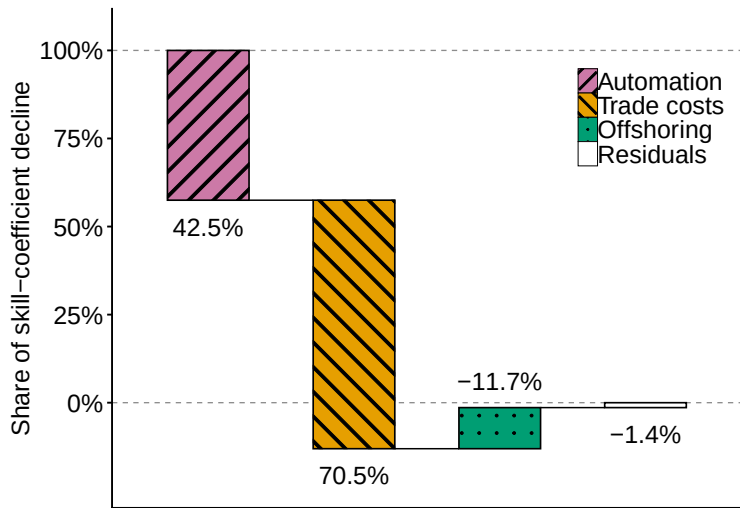
Offshoring to Advanced and Developing Economies



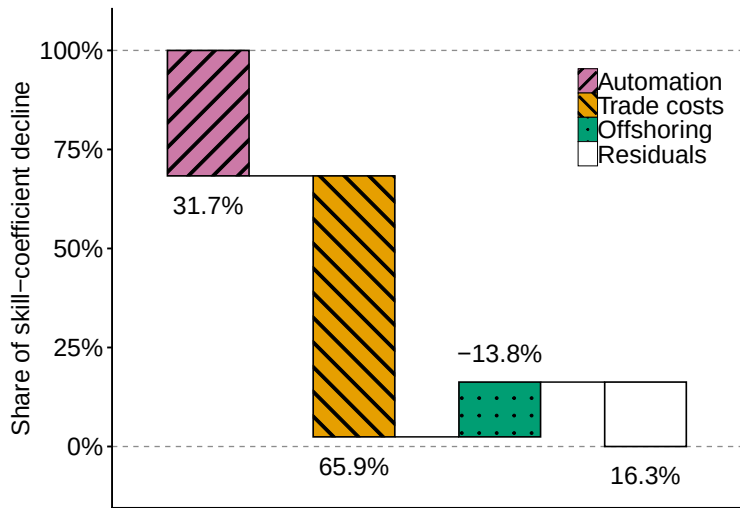
Adding Endowment and Deficit Changes



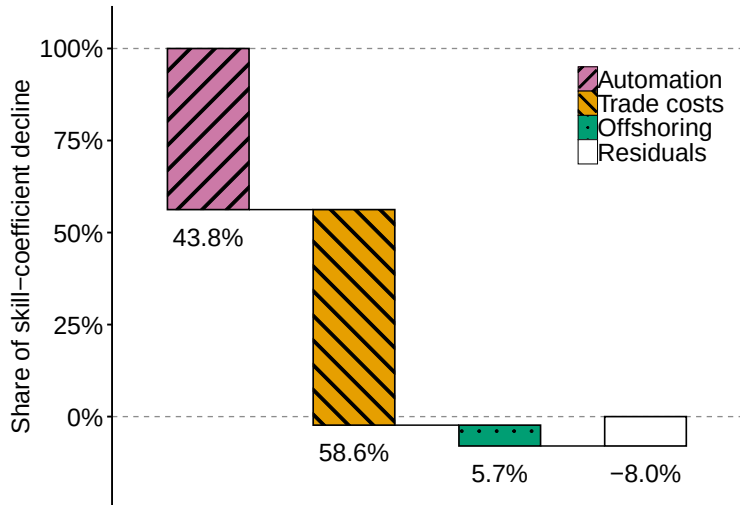
Weighted Decomposition: 2015 Exporter-Sector Exports



Alternative Task-Assignment Elasticity: $\varepsilon = 1.0$



Shorter Calibration Window: 1995–2014



Exact Hat Algebra: Tasks

- **Task assignment:** $\varepsilon = 1.4$; substitution elasticity $1 + \varepsilon = 2.4$
 - $\Gamma_{is}^{K,L}$: capital/labor task shares; Ω'_{is} : updated capital/labor odds

$$\Omega'_{is} = \frac{\Gamma_{is}^K}{\Gamma_{is}^L} \left(\frac{\widehat{w}_i^L}{\widehat{w}_{is}^K} \right)^\varepsilon, \quad \Gamma_{is}^{K'} = \frac{\Omega'_{is}}{1 + \Omega'_{is}}$$

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- **Task price** (ε : odds elasticity to w_i^L/w_{is}^K)

$$\widehat{w}_{is}^T = \widehat{w}_i^L \left(\frac{1 - \Gamma_{is}^{K'}}{1 - \Gamma_{is}^K} \right)^{1/\varepsilon}$$

Exact Hat Algebra: Sourcing

- **General- and intermediate-use prices**

- $\widehat{g}^G = \widehat{c}\widehat{\tau}$; $\widehat{g}^M = \widehat{c}\widehat{\tau}\widehat{\chi}$
- Initial sourcing support fixed

$$\widehat{p}_{jr}^u = \left[\sum_{i \in \mathcal{O}_{jr}^u} \pi_{ijr}^u \widehat{\lambda}_r (\widehat{g}_{ijr}^u)^{-\theta} \right]^{-1/\theta}$$

Channels: China and Other Trade Costs

Channel	China	Other trade costs	Joint
Market access	-0.99	-0.99	-0.98
Unit cost	+0.32	-0.32	-0.19
Expenditure	+0.02	+0.10	+0.04
Total	-0.66	-1.21	-1.14

- Market access accounts for most of the joint change
- Unit-cost effects have opposite signs

Welfare: Real Income

- **Representative-household real income**

- I_i : income; P_i^C : consumption price index

$$W_i = \frac{I_i}{P_i^C}$$

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- **Consumption price index**

- $v_{ir,1980}$: fixed expenditure shares; P_{ir}^G : general-use prices

$$\hat{P}_i^C = \exp \left(\sum_r v_{ir,1980} \ln \hat{P}_{ir}^G \right)$$